COL703|7203

Let $\phi = \forall y_1 \ldots \forall y_n \exists x \, \phi'$

Given a function symbol of arity n that does not occur in ϕ'

$$\psi = \forall y_1 \forall y_2 \ldots \forall y_n \, \phi' \left[f(y_1 \ldots y_n) / x \right]$$

ϕ and ψ are equisatisfiable.

Universe = $\{p, q, r\}$

Red = $\{p\}$

Adj = $\{(p,q), (q,r), (r,p), (r,q)\}$

$\forall x \exists y \left(Adj(x,y) \wedge \neg Red(y) \right) \wedge \exists z \, Red(z)$

$\forall x \left(Adj(x, f(x)) \wedge \neg Red(f(x)) \right) \wedge Red(c)$

$c \mapsto p$ / q

$f(p) = q$

$f(q) = r$

$f(r) = q$

$\forall x \left\{ Adj(x, f(x)) \wedge \neg Red(f(x)) \wedge Red(c) \right)$

→ Therefore not logically equivalent.

Herbrand structure

Let σ be a signature (with at least one constant symbol).

A σ-structure is called a Herbrand structure if

i) the universe U_H is the set of ground terms over σ (variable-free terms)

ii) for every constant symbol c $\quad c_H = c$

iii) for every k-ary function symbol f in σ and all ground terms $t_1 \dots t_k$ $\quad f_H(t_1 \dots t_k) = f(t_1 \dots t_k)$.

f_1 $\{ \{P(a), \neg P(b), Q(z)\}, \{\neg P(b), \neg Q(z)\} \}$

$H_{f_1} = \{a, b\}$

f_2 $\{ \{\neg P(x, f(y))\}, \{P(w, g(z))\} \}$

$H_{f_2} = \{ a, f(a), g(a), f(f(a)), f(g(a)), g(f(a)), g(g(a)), g(f(f(a))), \ldots \}$

$F_3 \quad \{ \{ \neg P(a, f(x,y)) \}, \{ P(b, f(x,y)) \} \}$

$H_{F_3} = \{ a, b, f(a,a), f(a,b), f(b,a), f(b,b), f(a, f(a,a)) \}$

Herbrand's Theorem

Let $\phi = \forall x_1 \ldots \forall x_n \phi'$ be a closed formula in Skolem form.

Then ϕ is satisfiable iff it has a Herbrand model/assignment.

$$\forall x \Big(P(x) \to Q(f(x)) \;\wedge\; P(a) \wedge \neg Q(a) \Big)$$

Universe = $\{0,1\}$

$P = \{0\}$

$Q = \{1\}$

$f(0) = 1 \;;\; f(1) = 0$

$a \mapsto 0$

Universe

$a \mapsto a$

$f(a) \mapsto f(a)$

$f(f(a)) \mapsto f(f(a))$

a , $f(a)$, $f(f(a))$, $f(f(f(a)))$, ..

P, Q ; $\neg P, Q$; $P, \neg Q$; $\neg P\, Q$

What about the interpretation for P and Q ?

$P = \{ f^K(a) \;|\; K \in \text{even} \}$

$Q = \{ f^K(a) \;|\; K \in \text{odd} \}$

$$\forall x \forall y \Big[\big(E(x,y) \rightarrow E(y,x) \big) \wedge E(x, S(x)) \wedge \neg E(c,c) \Big]$$

Universe $= \{0, 1, 2\}$

$S(0) = 1 \; ; \; S(1) = 0 \; ; \; S(2) = 2$

$c \mapsto 0$

$E = \{ (0,1), (1,0), (2,2) \}$

$c, \; S(c), \; S(S(c)), \; S(S(S(c))) \; \ldots$ ✓

$E\big(S^k(c), S^{k'}(c) \big)$ ✓

$= T$ if $k + k'$ is odd.

$x \quad S^m(c)$

$y \quad S^n(c)$

Is this satisfiable?

$$\exists x_1 \exists x_2 \exists x_3 \Big(\neg\big(\neg P(x_1) \to P(x_2)\big) \wedge \neg\big(\neg P(x_1) \to \neg P(x_3)\big) \Big)$$

$$\neg\big(\neg P(a) \to P(b)\big) \wedge \neg\big(\neg P(a) \to \neg P(c)\big)$$

P(a)	P(b)	P(c)
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Exercise: check

There may not always be a finite witness of satisfiability.

But if the formula is unsat, then there is always a finite proof.

$$\neg\Big(\forall x\,(P(x)\to Q(x)) \to (\forall x\,P(x) \to \forall x\,Q(x))\Big)$$

$$\neg\Big(\forall x\,(P(x)\to Q(x)) \to (\forall y\,P(y) \to \forall z\,Q(z))\Big)$$

$$\forall x\,(P(x)\to Q(x)) \wedge \neg(\forall y\,P(y) \to \forall z\,Q(z))$$

$$\wedge\ \forall y\,P(y) \wedge \exists z\,\neg Q(z)$$

$$\forall x\,\forall y\,(P(x)\to Q(x) \wedge P(y) \wedge \neg Q(c))$$

$$\forall x \forall y \left((P(x) \to Q(x)) \wedge P(y) \wedge \neg Q(c) \right)$$

$P(c) \to Q(c)$ $\qquad$ $P(c)$ $\qquad$ $\neg Q(c)$

$\neg P(c) \vee Q(c)$ $\qquad$ $P(c)$ $\qquad$ $\neg Q(c)$

$Q(c)$

$\square$